LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Wire Substitution for Repairs and Maintenance
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Wire Substitution for Repairs and Maintenance.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of wire stripping, insulation removal, and conductor matching using dummy wire samples; Safety: Ensure no live circuits, wear eye protection; Supervision: Instructor verifies gauge and material match before trainee proceeds.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary causes of aircraft wiring degradation?
(b) How does wire substitution impact overall aircraft electrical integrity?
(c) Why is matching wire gauge critical during repair procedures?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of wire substitution • Standards for wire replacement • Material compatibility requirements • Insulation type selection • Conductor strand matching	L & Ds	PPT, IB	15 min
(b)	• Identification of wire degradation • Selection of replacement wire • Stripping and preparation techniques • Termination methods for repairs • Insulation restoration procedures	L & Ds	PPT, PDF	20 min
(c)	• Quality inspection criteria • Documentation of repairs • Common errors in substitution • Preventive maintenance measures • Safety checks post-repair	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between wire gauge and current carrying capacity in substitution.
(b) Describe the steps required to ensure a repaired wire meets safety standards.

12. Home Assignment and Follow Up	Prepare a brief report on the hazards of improper wire routing and protection, linking findings to the next lesson plan (LP 09) on wire routing hazards.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation on Wire routing and protection hazards.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min